MindMetrix Home Assignment – Biomarker Data Scientist

This is a home assignment to assess how you approach problems: especially, how you conduct, implement, and explain a data analytics task with a synthetic data set mimicking real-world data, while focusing on reproducibility and a clean architecture of your choice. This assignment will be followed by a technical interview. Please aim to spend no more than 4h on the assignment.

Data Set

The data set captures participants performing 10 trials of a relaxation exercise. Each trial consists of: 5s baseline, 30s relaxation, 5s break. During this relaxation exercise, several physiological signals are captured through a VR-based eye-tracking system. Assume that light conditions are constant.

The aim is to establish a relationship between state markers/metadata of the participants and physiological markers of the time series.

The data set consists of two .csv files:

(1) Subjects.csv: participant metadata.
(2) Timeseries.csv: time series data of physiological signals.

Assignment

Your task is to implement a reproducible preprocessing pipeline suitable for physiological data. Afterwards, explore and create meaningful inter- or intra-participant features and their relationship with participant metadata. This can involve clustering of participants.

Finally, use at least one exploratory unsupervised or dimensionality reduction method on your engineered features. Clearly state your hypotheses and justify your feature engineering decisions based on physiological reasoning.

How would you prove that the detected (bio)markers are sensitive?

Focus areas

- Feature design thinking & signal interpretation
- Physiological reasoning
- Reliable data analytics
- Presentation/visualization of results
- Code structure and clarity

Optional

Describe how you would connect the data pipeline to a backend API.

Tech Guidelines

- Use Python for implementation.

- Keep it simple but clean, with consistent naming and structure.
- Focus on demonstrating clarity, reliability, and maintainable architecture.
- Show awareness of scalability and future integration with backend systems.
- Write down your assumptions if something is left open or ambiguous.
- Specify any external libraries used and justify non-standard choices.

Deliverables

- Notebook, Git Repository or ZIP file that runs end-to-end from the raw CSV files

- Short report (max 1 page) covering your approach to the problem, explaining processing steps, model results and interpretations, as well as limitations and what you would do with more time. Additionally, quickly describe how you allocated the time.

Interview Discussion

We will talk about how you approach such problems, about the workflow, and the result. You may also describe how this code could integrate into our backend and how you think this could scale and be updated as more user data is collected. In addition, we will discuss how to handle code freezes and version control under a regulated process.

Expected effort

~3.5 hours implementation + 30 min documentation. The focus is on structured thinking and sound reasoning, not on perfect models.

Submission

Send a GitHub link, Notebook, or ZIP archive to job@mindmetrix.ch with subject: "[Your Name] - Assignment – Biomarker Data Scientist"